Generic Programming

Advanced Programming

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Introduction

“Generic programming centers around the idea of abstracting from concrete algorithms to obtain generic algorithms that can be combined with different data representations to produce a wide variety of useful software.”

— Musser, David R. & Stepanov, Alexander A.

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"Generic programming centers around the idea of abstracting from concrete algorithms to obtain generic algorithms that can be combined with different data representations to produce a wide variety of useful software."

— Musser, David R. & Stepanov, Alexander A.

"Lift algorithms and data structures from concrete examples to their most general and abstract form."

— Bjarne Stroustrup

Introduction

```
1.  int sum(int a, int b){
2.      return a + b;
3.  }
4.
5.  double sum(double a, double b){
6.      return a + b;
7.  }
8.
9.  long sum(long a, long b){
10.     return a + b;
11. }
```

“concrete algorithms”

Introduction

```
1.  int sum(int a, int b){
2.      return a + b;
3.  }
4.
5.  double sum(double a, double b){
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7.  }
8.
9.  long sum(long a, long b){
10.     return a + b;
11. }
```

"concrete algorithms"

Replace

```
1.  Number sum(Number a, Number b){
2.      return a + b;
3.  }
```

"generic algorithms"

Templates

```
1.  int sum(int a, int b){
2.      return a + b;
3.  }
4.
5.  double sum(double a, double b){
6.      return a + b;
7.  }
8.
9.  long sum(long a, long b){
10.     return a + b;
11. }
```

"concrete algorithms"

Generate*

```
1.  template <typename Number>
2.  Number sum(Number a, Number b){
3.      return a + b;
4.  }
```

"generic algorithms"

*at compile-time

Templates

```
1.  template <typename Container>
2.  void printContainer(Container& container) {
3.    for (auto element : container)
4.        cout << element << endl;
5.  }
6.
7.  int main() {
8.     vector<int> v{1,2,3,4,5};
9.     printContainer(v);
10. }
```

"generic algorithms"

Function Templates

```
1.  template <typename Number>
2.  Number sum(Number a, Number b){
3.      return a + b;
4.  }
```

```
1.  template <typename T1, typename T2>
2.  void print(T1 a, T2 b){
3.      cout << a << " " b << endl;
4.  }
```

Function Templates

```
1.  template <typename Number>
2.  Number sum(Number a, Number b){
3.      return a + b;
4.  }
```

```
1.  template <typename T1, typename T2>
2.  void print(T1 a, T2 b){
3.      cout << a << " " b << endl;
4.  }
```

```
1.  cout << sum(5, 8) << endl;
2.  cout << sum(5.0, 8.5) << endl;
3.
4.  print(5, 8);
5.  print(5.0, "Hello");
```

Template Instantiation

```
1.  template <typename Number>
2.  Number sum(Number a, Number b){
3.      return a + b;
4.  }
```

The compiler generates no binary code from a source file with only template definitions.

→ Templates must be instantiated in order for code to be generated!

Template Instantiation

```
1.  template <typename Number>
2.  Number sum(Number a, Number b){
3.      return a + b;
4.  }
```

Implicit Template Instantiation:
When calling a template function, it is implicitly instantiated with the called type.
⟶ Code is generated for this function.

```
1.  cout << sum(5, 8) << endl;
2.  cout << sum(5.0, 8.5) << endl;
```

<u>Instantiates and generates</u>:
int sum(int, int)
double sum(double, double)

Template Instantiation

```
1.  template <typename Number>
2.  Number sum(Number a, Number b){
3.      return a + b;
4.  }
```

Explicit Template Instantiation:
An explicit instantiation definition forces instantiation of the function.
⟶ Code is generated for this function.

```
1.  template int sum(int, int);
2.  template double sum(double, double);
```

<u>Instantiates and generates</u>:
int sum(int, int)
double sum(double, double)

Template Argument Deduction

```
1.  template <typename Number>
2.  Number sum(Number a, Number b){
3.      return a + b;
4.  }
```

```
1.  // compilation error:
2.  // no matching function for call to 'sum(int, double)'
3.  cout << sum(5, 8.5) << endl;
```

The template type must be deducible at compile-time based on the argument types.

→ **If not possible: compiler error!**

Template Argument Deduction

```
1.  template <typename Number>
2.  Number sum(Number a, Number b){
3.      return a + b;
4.  }
```

```
1.  // compilation error:
2.  // no matching function for call to 'sum(int, double)'
3.  cout << sum(5, 8.5) << endl;
4.
5.  // implicit conversion from int to double
6.  cout << sum<double>(5, 8.5) << endl;
```

The template type must be deducible at compile-time based on the argument types.

→ **If not possible: compiler error!**

You can bypass template argument deduction by providing the template type explicitly.

Template Argument Deduction

```
1.  template int sum<int>(int, int); // explicit template type
2.  template int sum<>(int, int); // template type deduced
3.  template int sum(int, int); // template type deduced
4.
5.  int main() {
6.    cout << sum<int>(5, 8) << endl; // explicit template type
7.    cout << sum<>(5, 8) << endl;    // template type deduced
8.    cout << sum(5, 8) << endl;      // template type deduced
9.  }
```

You can bypass template argument deduction by providing the template type explicitly.

→ **Both in implicit and explicit instantiations!**

Explicit Specialization of a Template Function

```
1.  template <typename Number>
2.  Number sum(Number a, Number b){
3.    return a + b;
4.  }
5.
6.  template <>
7.  double sum<double>(double a, double b){
8.    int rounded = a + b;
9.    return rounded ;
10. }
11.
12. int main() {
13.   cout << sum(5, 8) << endl;      // generic version
14.   cout << sum(5.0, 8.5) << endl; // specific version
15. }
```

You can overload the general version of the template function with a specific implementation.

Example:
vector<TElement> vs vector<bool>

If constexpr

```
1.  template<typename Number>
2.  void sum(Number a, Number b) {
3.     if constexpr (std::is_same<Number,double>::value) {
4.          int rounded = a + b;
5.          return rounded ;
6.     } else {
7.          return a + b;
8.  }}
```

Evaluate if statement at compile-time.

→ Generated functions will not contain the if.

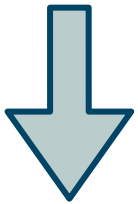
Template Function Overloading

```
1.  template <typename Number>
2.  Number sum(Number a, Number b){
3.    return a + b;
4.  }
5.
6.  template <typename Number>
7.  Number sum(Number a, Number b, Number c){
8.    return a + b + c;
9.  }
10.
11. int main() {
12.   cout << sum(5, 8) << endl;
13.   cout << sum(5, 8, 2) << endl;
14. }
```

Template functions can be overloaded similar to non-template functions.

Abbreviated Template Function (since c++20)

```
1.  void print(auto a, auto b) {
2.      cout << a << " " b << endl;
3.  }
```



```
1.  template <typename T1, typename T2>
2.  void print(T1 a, T2 b) {
3.     cout << a << " " b << endl;
4.  }
```

Template functions can be abbreviated with auto parameters.

→ Translated to template function by compiler.

auto keyword (since c++11)

The **auto** keyword allows for automatic type-deduction at compile-time.

→ **Benefits?**

```
1.  auto x = 42;
```

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The **auto** keyword allows for automatic type-deduction at compile-time.

```
1.    auto x = 42;
```

→ **Benefits:**

- Correctness: auto guarantees the correct type is used.

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The **auto** keyword allows for automatic type-deduction at compile-time.

```
1.    auto x = 42;
```

→ **Benefits:**

- Correctness: auto guarantees the correct type is used.
- Maintainability and robustness: When an expression's type changes, auto will continue to resolve correctly.

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```
1.    auto x = 42;
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→ **Benefits:**

- Correctness: auto guarantees the correct type is used.
- Maintainability and robustness: When an expression's type changes, auto will continue to resolve correctly.
- Performance: auto guarantees no implicit conversions.

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```
1.    auto x = 42;
```

→ **Benefits:**

- Correctness: auto guarantees the correct type is used.
- Maintainability and robustness: When an expression's type changes, auto will continue to resolve correctly.
- Performance: auto guarantees no implicit conversions.
- Usability: easier to use for long or complex types (such as function pointers)

auto keyword (since c++11)

"write code against interfaces, not implementations"

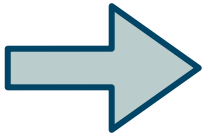
```
1.  set<string> getData() {
2.      return {"one", "two", "three"};
3.  }
```

```
1.  vector<int> getData() {
2.      return {1, 2, 3};
3.  }
```

```
1.  int main() {
2.      auto data = getData();
3.      for (auto element : data)
4.          cout << element << endl;
5.  }
```

Trailing Return Type (since c++11)

```
1.  vector<int> getData() {
2.      return {1, 2, 3};
3.  }
```



```
1.  auto getData() -> vector<int> {
2.      return {1, 2, 3};
3.  }
```

Trailing Return Type (since c++11)

- Can use parameters as part of the return type:

```
1.  template <typename Number>
2.  auto sum(Number a, Number b) -> decltype(a + b) {
3.    return a + b;
4.  }
```

Trailing Return Type (since c++11)

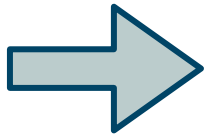
- Can use parameters as part of the return type.
- Can access class (private) type aliases without namespace specifier:

```
1.  class Rectangle {
2.    private:
3.        using Corners = vector<Coordinate>;
4.        Corners _corners;
5.    public:
6.        auto getCorners() -> Corners;
7.  };
8.
9.  auto Rectangle::getCorners() -> Corners {
10.   return _corners;
11. };
```

Trailing Return Type (since c++11)

- Can use parameters as part of the return type.
- Can access class type aliases without namespace specifier.
- Easy to switch to automatic return type deduction in the future*:

```
1.  auto getData() -> vector<int> {
2.      return {1, 2, 3};
3.  }
```

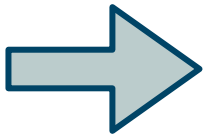


```
1.  auto getData() {
2.      return vector<int>{1, 2, 3};
3.  }
```

*automatic return type deduction (since c++14)

Automatic Return Type Deduction (since c++14)

```
1.  auto getData() -> vector<int> {
2.      return {1, 2, 3};
3.  }
```



```
1.  auto getData() {
2.      return vector<int>{1, 2, 3};
3.  }
```

Function return types can be deduced at compile-time.

Class Templates

```
1.  template <typename Element>
2.  class Queue {
3.      private:
4.          vector<Element> _elements;
5.      public:
6.          void push(const Element& e);
7.          Element front();
8.  };
```

Class Templates

```
1.  template <typename Element>
2.  class Queue {
3.    private:
4.        vector<Element> _elements;
5.    public:
6.        void push(const Element& e);
7.        Element front();
8.  };
```

```
1.  template <typename Element>
2.  void Queue<Element>::push(const Element& e) {
3.    _elements.push_back(e);
4.  }
```

Class Templates

```
1.  template <typename Element>
2.  class Queue {
3.    private:
4.      vector<Element> _elements;
5.    public:
6.      void push(const Element& e);
7.      Element front();
8.  };
```

```
1.  template <typename Element>
2.  void Queue<Element>::push(const Element& e) {
3.    _elements.push_back(e);
4.  }
```

The compiler generates no binary code from a source file with only template definitions.

→ Templates must be instantiated in order for code to be generated!

Class Templates

```
1.  template <typename Element>
2.  class Queue {
3.    private:
4.        vector<Element> _elements;
5.    public:
6.        void push(const Element& e);
7.        Element front();
8.  };
```

```
1.  int main() {
2.      Queue<int> q;
3.      q.push(1);
4.      q.push(2);
5.      q.push(3);
6.      cout << q.front() << endl;
7.  }
```

The compiler generates no binary code from a source file with only template definitions.

⟶ Templates must be instantiated in order for code to be generated!

Class Templates

```
1.  template<typename FirstType, SecondType>
2.  class MyPair {
3.      FirstType _first;
4.      SecondType _second;
5.    public:
6.      MyPair(FirstType a, SecondType b) : _first{a}, _second{b} {};
7.
8.      inline void print() const {
9.          cout << _first << ", " << _second << endl;
10.     }
11. };
```

```
1.  MyPair<int, string> p{5, "Hello"};
2.  p.print();
```

Class Template Argument Deduction (since C++17)

```
1.  template<typename FirstType, SecondType>
2.  class MyPair {
3.      FirstType _first;
4.      SecondType _second;
5.    public:
6.      MyPair(FirstType a, SecondType b) : _first{a}, _second{b} {};
7.
8.      inline void print() const {
9.          cout << _first << ", " << _second << endl;
10.     }
11. };
```

```
1.  MyPair p{5, "Hello"};
2.  p.print();
```

Default Template Arguments

```
1.  template<typename FirstType = int, SecondType = int>
2.  class MyPair {
3.      FirstType _first;
4.      SecondType _second;
5.    public:
6.      MyPair(FirstType a, SecondType b) : _first{a}, _second{b} {};
7.
8.      inline void print() const {
9.          cout << _first << ", " << _second << endl;
10.     }
11. };
```

```
1.  short x = 5, y = 9;
2.  MyPair p{x, y}; // implicit conversion short -> int
3.  p.print();
```

Templates & Inheritance

```
1.  class IReporter {
2.    public:
3.      // compilation error
4.      template<typename StringType>
5.      virtual void print(StringType message) = 0;
6.  };
```

Template functions cannot be virtual.

Templates & Inheritance

```
1.  class IReporter {
2.    public:
3.      // compilation error
4.      template<typename StringType>
5.      virtual void print(StringType message) = 0;
6.  };
```

Template functions cannot be virtual.

→ Templated functions provide compile-time polymorphism, virtual functions provide runtime polymorphism.

Templates & Inheritance

```
1.  class IReporter {
2.    public:
3.      // compilation error
4.      template<typename StringType>
5.      virtual void print(StringType message) = 0;
6.  };
```

Template functions cannot be virtual.

→ Templated functions provide compile-time polymorphism, virtual functions provide runtime polymorphism.

(Size of the vtable would not be known until the entire program has been translated.)

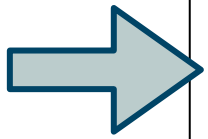
Templates & Inheritance

```
1.  template<typename StringType>
2.  class IReporter {
3.    public:
4.      virtual void print(StringType message) = 0;
5.  };
6.
7.  template<typename StringType>
8.  class Reporter : public IReporter<StringType>{
9.    public:
10.     void print(StringType message) override;
11. };
```

A template class can contain virtual functions!

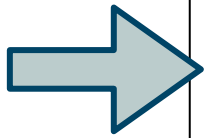
Templates & Inheritance

```
1.  class IReporter {
2.    public:
3.      // compilation error
4.      template<typename StringType>
5.      virtual void print(StringType message) = 0;
6.  };
```



One class with unknown amount of overloaded print functions.

```
1.  template<typename StringType>
2.  class IReporter {
3.    public:
4.      virtual void print(StringType message) = 0;
5.  };
```



Unknown amount of classes with one print function each.

Universal Reference

```
1.  template <typename T>
2.  void func(T && t);
```

A **universal reference** is an rvalue reference to a template parameter, which can then be deduced as either a rvalue or an lvalue reference.

Universal Reference

```
1.  template <typename T>
2.  void func(T && t);
```

A **universal reference** is an rvalue reference to a template parameter, which can then be deduced as either a rvalue or an lvalue reference.

→ Also called a **forwarding reference**.

std::forward (since C++14)

```
1.  struct A {
2.   A(int&& n) { cout << "rvalue A"; }
3.   A(int& n)   { cout << "lvalue A"; }
4.  };
5.
6.  template <typename Number>
7.  A AFactory(Number&& n){
8.    return A{std::forward<Number>(n)};
9.  }
10.
11. int main() {
12.   int x = 1;
13.   AFactory(1); // rvalue A
14.   AFactory(x); // lvalue A
15. }
```

std::forward preserves the value category of its argument.

std::forward (since C++14)

```
1.  struct A {
2.   A(int&& n) { cout << "rvalue A"; }
3.   A(int& n)   { cout << "lvalue A"; }
4.  };
5.
6.  template <typename Number>
7.  A AFactory(Number&& n){
8.    return A{std::forward<Number>(n)};
9.  }
10.
11. int main() {
12.   int x = 1;
13.   AFactory(1); // rvalue A
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15. }
```

std::forward preserves the value category of its argument.

perfect forwarding: objects passed as lvalue are copied, and objects passed as rvalue are moved.

std::forward (since C++14)

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1.  struct A {
2.   A(int&& n) { cout << "rvalue A"; }
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4.  };
5.
6.  template <typename Number>
7.  A AFactory(Number&& n){
8.    return A{std::forward<Number>(n)};
9.  }
10.
11. int main() {
12.   int x = 1;
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14.   AFactory(x); // lvalue A
15. }
```

std::forward preserves the value category of its argument.

perfect forwarding: objects passed as lvalue are copied, and objects passed as rvalue are moved.

⟶ avoids excessive copying, and avoids having to write multiple overloads for lvalue and rvalue references.

Parameter pack (since C++11)

A template **parameter pack** is a template parameter that accepts zero or more template arguments.

Syntax: template <typename… PackName>

A function **parameter pack** is a function parameter that accepts zero or more function arguments.

Syntax: functionName(PackName… parameterName)

Parameter pack (since C++11)

```
1.  template <typename Arg, typename...Args>
2.  void print(Arg&& arg, Args&&... args){
3.     cout << arg;
4.     ((cout << ' ' << args), ...);
5.  }
6.
7.  int main() {
8.     print("Grade: ", 15,5, '/', 20);
9.  }
```

Parameter pack (since C++11)

```
1.  template <typename Arg, typename...Args>
2.  void print(Arg&& arg, Args&&... args){
3.      cout << arg;
4.      ((cout << ' ' << args), ...);
5.  }
6.
7.  int main() {
8.      print("Grade: ", 15.5, '/', 20);
9.  }
```

template parameter pack

(0 or more types)

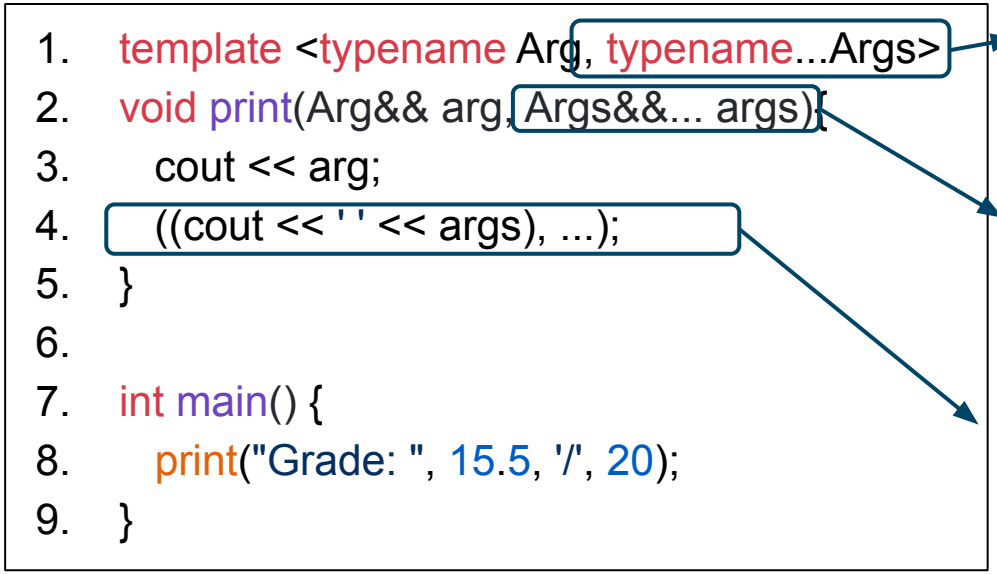
Parameter pack (since C++11)

```
1.  template <typename Arg, typename...Args>
2.  void print(Arg&& arg, Args&&... args){
3.      cout << arg;
4.      ((cout << ' ' << args), ...);
5.  }
6.
7.  int main() {
8.      print("Grade: ", 15.5, '/', 20);
9.  }
```

template parameter pack
(0 or more types)

function parameter pack
(0 or more parameters)

Parameter pack (since C++11)



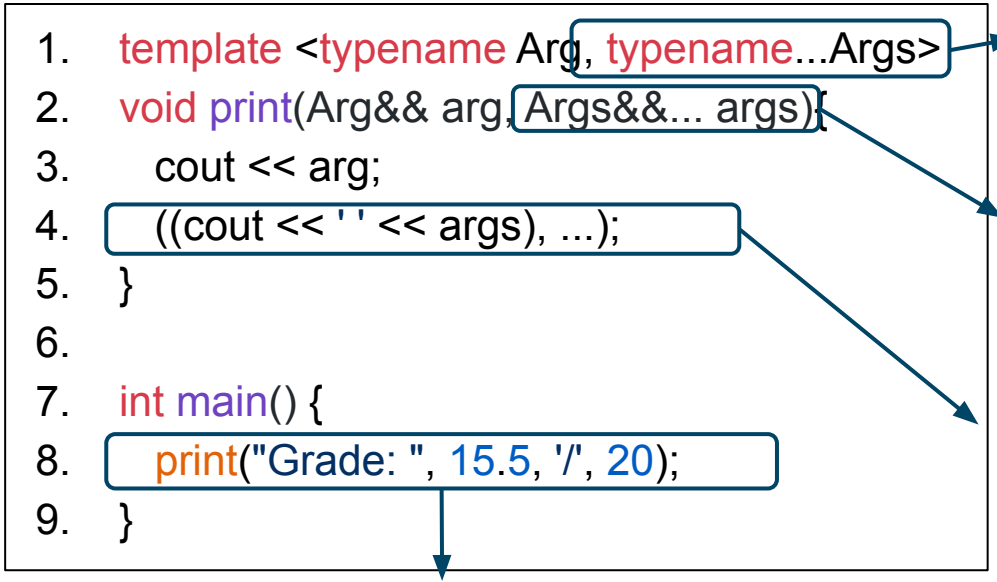
```
1.  template <typename Arg, typename...Args>
2.  void print(Arg&& arg, Args&&... args){
3.     cout << arg;
4.     ((cout << ' ' << args), ...);
5.  }
6.
7.  int main() {
8.     print("Grade: ", 15.5, '/', 20);
9.  }
```

template parameter pack
(0 or more types)

function parameter pack
(0 or more parameters)

fold expression: Reduces a parameter pack over an operator. (since C++17)

Parameter pack (since C++11)



template parameter pack
(0 or more types)

function parameter pack
(0 or more parameters)

fold expression: Reduces a parameter pack over an operator. (since C++17)

Function can be called with any number of parameters, with any number of types!

Fold Expressions (since C++17)

A fold expression expands an expression with *n* parameters as follows:

- Unary Right Fold:

  (E op …) becomes (E_1 op (... op (E_{n-1} op E_n)))

  ({1,2,3,4} + …) becomes (1 + (2 + (3 + 4)))

Fold Expressions (since C++17)

A fold expression expands an expression with *n* parameters as follows:

- Unary Right Fold:

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  (... op E) becomes (((E_1 op E_2) op …) op E_n)

Fold Expressions (since C++17)

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- Unary Left Fold:
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- Binary Right Fold:
  (E op … op L) becomes (E_1 op (... op (E_{n-1} op (E_n op L))))
- Binary Left Fold:
  (L op … op E) becomes ((((L op E_1) op E_2) op …) op E_n)